

HELPS: Helicopter Emergency Landing Place Suggestion Systems

Literature Review Submission on

Algorithmic Foundations, Terrain Perception, Geometric Constraints, and Regulatory Frameworks for Autonomous

Abstract This combined review synthesizes thirty three peer reviewed studies underpinning the Helicopter Emergency Landing Place Suggestion System (HELPSs). It unifies two previously independent reviews into a single body of literature spanning satellite and vision based obstacle detection, land cover classification, deterministic and GPU accelerated Connected Component Analysis (CCA), macro landform and terrain classification, terrain representation, digital elevation model (DEM) construction and error propagation, slope and aspect derivation, 3D landing site candidate extraction and validation, Pole of Inaccessibility (PoI) and maximal clearance geometry, autorotation trajectory optimization, international heliport design mandates, degraded visual environment (DVE) aeromechanics, and 3D communication aware airspace geofencing. Each domain is mapped onto the eight modules of the HELPSs Terrain Knowledge Generation and Helicopter Landing Recommendation pipelines defined in the System Design Document (SDD), and every paper is flagged by priority against the module it supports. The review closes by identifying the gaps that remain open across the combined literature, most notably the absence of any published work coupling real time terrain and land cover perception to autorotation dynamics optimal control.

Index Terms Autonomous helicopters, emergency landing systems, autorotation, computer vision, connected component analysis, computational geometry, Pole of Inaccessibility, digital elevation models, slope analysis, land cover classification, terrain representation, aviation regulations, degraded visual environments.

I. INTRODUCTION

Autonomous rotorcraft emergency landing depends on two problems that have historically been studied in separate literatures: whether a candidate site is physically and geometrically safe to occupy, and whether the aircraft's flight dynamics permit a controlled descent to that site at all. Approximately half of all helicopter accidents are attributed to engine failure, and autorotation is the only unpowered recovery manoeuvre available to the crew. At the macro scale, candidate operational zones across broad geographic expanses are pre screened using optical satellite imagery, land cover classification, and Digital Elevation Models (DEMs). At the micro scale, onboard perception verifies slope, roughness, footprint clearance, and the absence of localized obstructions during final descent. This review merges twenty one papers on the algorithmic and regulatory side of this problem with twelve papers on terrain perception, DEM construction, and autorotation dynamics, producing a single literature base of thirty three studies organized around the modules of the HELPSs SDD.

Section II covers vision and satellite based obstacle detection. Section III covers land cover classification. Section IV covers Connected Component Analysis. Section V covers macro landform and terrain classification. Section VI covers terrain representation. Section VII covers DEM construction and error propagation. Section VIII covers slope and aspect derivation. Section IX covers landing zone detection and validation. Section X covers Pole of Inaccessibility and geometric clearance. Section XI covers autorotation trajectory optimization. Section XII covers aviation regulations and heliport design. Section XIII covers operable weather and degraded visual environments. Section XIV covers restricted airspace and communication constraints. Section XV synthesizes all thirty three papers against the SDD module structure and identifies remaining research gaps, and Section XVI concludes.

II. OBSTACLE DETECTION VIA SATELLITE AND VISION SYSTEMS

A. Vision Based Obstacle Detection Frameworks

Badrloo et al. [1] surveyed over 110 studies published over two decades, establishing a comparative benchmark between monocular and stereoscopic vision architectures for autonomous vehicle navigation. Monocular frameworks favor appearance centric segmentation (Otsu thresholding, Markov Random Fields, YOLOv3/v4, Faster R CNN) and motion based features (optical flow, SIFT, SURF with SVM classifiers), offering low computational overhead but no direct 3D metric scale. Stereoscopic configurations use Inverse Perspective Mapping and Semi Global Matching to recover 3D point clouds directly but impose heavier compute and calibration demands. In HELPSs, offline satellite segmentation handles macro terrain screening (Modules 2/5), while lightweight monocular or stereo perception is reserved for the terminal descent phase.

B. Modified Deep Learning for Small Object Satellite Segmentation

Ren et al. [2] addressed the failure of standard detectors on small features (10^2 – 50^2 pixels) in satellite imagery by constructing a Fine Resolution Feature Pyramid over ResNet 50, reducing feature stride from 16 to 8 pixels, and replacing PASCAL VOC anchors with small object anchors (16^2 , 40^2 , 100^2). Reduced stride raised small target accuracy from 66.6% to 73.5% mAP, and contextual fusion with rotational augmentation pushed accuracy to 78.9% mAP. This ensures small structures are not misclassified as clear terrain during Module 2 raster binarization.

C. Non Deep Learning Urban and Building Extraction

Sırmaçek and Ünsalan [3] developed a deterministic pipeline for urban boundary and building footprint extraction from 1 meter Ikonos imagery using Bilateral Filtering, SIFT keypoint graphs, and Graph Cut segmentation, achieving 89.62% correct urban detection with an 8.03% false alarm rate and 88.4% building footprint extraction. This deterministic pipeline is an offline fallback for Modules 2–4 when training data for deep models is unavailable.

III. LAND COVER CLASSIFICATION

Vali, Comai, and Matteucci [4] reviewed deep learning approaches to land use and land cover (LULC) classification from hyperspectral and multispectral imagery, documenting rapid growth since 2015 and cataloging the major public benchmark datasets. The review establishes that convolutional, end to end feature learning has displaced hand engineered spectral indices as the dominant methodology, directly supporting the terrain classification step of Module 2.

Mountrakis and Heydari [5] built a labeled reference set of roughly 21 million annual temporal sequences from the Landsat 5/7/8 archive and compared deep classifiers against a Random Forest baseline. Random Forest achieved ~70% average accuracy versus 86%–95% for deep models, with the largest gains in historically difficult classes: grass and bare land rose from 63–65% to 79%–96%. Fusing two overlapping sensors improved F1 on difficult edge pixels by up to 15%, informing HELPSs' choice of classifier and multi sensor fusion strategy for terrain classification (Module 2, Step 1).

Hosseiny, Abdi, and Jamali [6] compared shallow and deep learners across 141 features, 85 scenarios, and 8 LULC classes for Sentinel 2 imagery of Malmö, finding mean accuracy rose from 84.64% to 92.63% with deep learners, and applied Shapley additive explanations for feature level interpretability. Feature level explainability of this kind is directly relevant to HELPSs' goal of explainable recommendations (Module 7 of the Recommendation Pipeline).

IV. CONNECTED COMPONENT ANALYSIS (CCA) ALGORITHMS

A. Linear Time Sequential Local CCA

Suzuki et al. [7] formulated a deterministic linear time labeling scheme using a compact 1D Label Connection Table to resolve equivalences without tree structures, labeling non convex regions in a maximum of four passes with scaling $t_{\max} = a \cdot I^{1.03}$, versus $I^{1.5}$ – $I^{1.9}$ for conventional methods. This provides guaranteed $O(N)$ foreground processing for multi gigabyte GeoTIFF rasters in Modules 2/4 and directly implements the Connected Terrain Formation and Connected Operational Zone Extraction steps of Modules 2 and 4.

B. Optimized 2×2 Block Scanning Two Scan CCA

He et al. [8] reduced redundant neighbor re checking by processing 2×2 pixel blocks concurrently with 16 configuration based decision rules, cutting per pixel neighbor evaluations by ~50% and achieving up to a 16% speedup, accelerating the eight threshold slope segmentation of Module 4 (Phase 8: System Optimization).

C. Massively Parallel GPU Accelerated CCA

Kalentev et al. [9] introduced an enhanced Parallel Label Equivalence algorithm (Extended Hoshen Kopelman) for NVIDIA CUDA, yielding 15×–35× speedups and labeling an 8128×8128 binary matrix in under 388 ms within three parallel iterations, giving HELPSs a GPU acceleration path for concurrent labeling and zone area computation as the system scales to higher resolution regions (Phase 8 Roadmap: Future Scalability).

V. MACRO LANDFORM AND TERRAIN CLASSIFICATION

Dal and Görüm [10] investigated how analysis window size and DEM resolution affect the definition of mountain–plateau and mountain–plain transitions using the UNEP WCMC K1 algorithm on Turkish terrain at multiple resolutions (GMTED2010, MERIT DEM). Their central finding is that topographic relief – the ratio of maximum to minimum elevation in a window – is more sensitive to window size than to elevation, slope, or DEM resolution itself, so relief computed at one window size cannot be compared directly to relief at another. Any HELPSs terrain scoring metric that uses relief must fix or normalize its analysis window and widen safety margins when only coarse DEM data is available (Module 3: Terrain Characterization).

Arrell et al. [11] applied fuzzy c means clustering to morphometric indices (curvature and slope derived elevation derivatives) at 50 m and 100 m resolution over Snowdonia, Wales. Convergent and divergent slope classes – the classes most likely to correspond to obstacles such as ridges and gullies reclassified substantially between resolutions, while flat, planar areas persisted as a stable, scale independent class. This indicates that HELPSs' flat site detection (candidate terrain identification) is comparatively robust to resolution choice, while obstacle class boundaries require multi resolution confirmation before being trusted.

VI. TERRAIN REPRESENTATION

Shi and Xue [12] addressed deriving the minimum set of viewpoints for maximum visibility coverage over a large DEM via automatic viewpoint generation, per viewpoint viewshed calculation, and NP hard combinatorial minimum set selection. Validated against the Summersville DEM, the method identified over 4,000 candidate viewpoints across roughly 16.8 million grid cells, requiring more than 64 GB of memory for full analysis – illustrating the computational cost that HELPSs must budget for if terrain visibility or approach corridor analysis (Module 5) is extended to exhaustive coverage.

Zheng, Xiong, Gong, and Yue [13] proposed a multi resolution triangulated irregular network (TIN) pyramid, built with a point additive algorithm constrained by extracted landscape features, for representing large scale terrain without losing morphological fidelity across levels of detail. Compared against conventional grid based bilinear resampling, the TIN approach preserved slope morphology and fine structures such as depressions and ridges, while the grid based method systematically raised depressions and flattened protuberances – an over smoothing effect that could mask or fabricate flat areas. This is directly relevant to how the Terrain Knowledge Database (Chapter 5 of the SDD) stores Operational Zone polygon geometry across multiple slope thresholds without losing fidelity.

VII. DIGITAL ELEVATION MODELS: CONSTRUCTION AND ERROR

Chen and Yue [14] developed a DEM construction method based on the theorem of surfaces (SMTS) and formally defined terrain representation error (ETR) as a component of total DEM error alongside sampling and interpolation error. SMTS outperformed IDW, SPLINE, and KRIGING in both numerical and real world tests, though the real world case showed larger accuracy loss, partly attributed to ETR affecting SMTS more than classical interpolators. The authors argue total DEM error, not sampling or interpolation error alone, should be the standard accuracy index – a consideration directly relevant to Module 1's DEM acquisition and standardization steps.

Aguilar et al. [15] developed a hybrid theoretical–empirical model for vertical error in LiDAR derived DEMs of non open terrain, decomposing error into sample data error and gridding/interpolation error (inverse distance weighting, five nearest neighbors), calibrated against 29 sites in Almeria, Spain. Total error was insensitive to neighbor count within the typical four to ten range but sensitive to terrain slope and sample density – meaning HELPSs' DEM derived flatness estimates are less trustworthy over steep or sparsely sampled terrain, reinforcing the need for conservative slope threshold margins in Module 4.

VIII. SLOPE AND ASPECT DERIVATION

Jones [16] systematically compared eight 3×3 kernel algorithms for computing slope and aspect from a DEM, benchmarked against Morrison's Surface III, a synthetic surface with analytically known gradient and aspect. Fleming and Hoffer's method produced the lowest error on the synthetic surface, Horn's method performed well on both synthetic and real DEMs, and the Maximum Downward Gradient method consistently ranked worst – a direct input to selecting the slope algorithm used in Module 4's Terrain Slope Computation step.

Zhou and Liu [17] investigated how DEM grid resolution, elevation precision, and grid orientation introduce systematic error into derived slope and aspect using two mathematically defined synthetic surfaces. Third order finite difference algorithms were considerably more sensitive to grid orientation than second order or simple difference algorithms, and all algorithms exhibited a recurring octant bias with error peaks every 45° of orientation. This implies a marginal slope candidate Operational Zone could be accepted or rejected purely as a function of DEM grid orientation – a risk HELPSs' multi threshold slope segmentation (Module 4, Step 2) should account for with margin rather than a hard cutoff.

IX. LANDING ZONE DETECTION AND VALIDATION

A. Full Scale Autonomous Helicopter Landing Systems

Scherer et al. [18] validated full scale autonomous landing in unprepared environments using Boeing Unmanned Little Bird and Eurocopter EC 135 platforms with a two tier evaluation: a Coarse Evaluation using a scrolling hash grid with moment based least squares plane fitting and Welford's algorithm for altitude variance filtering (excluding slopes over 5°), and a Fine Evaluation applying Delaunay triangulation against a 3D structural aircraft model to simulate skid touchdown, attitude limits, and clearance volumes. Flight corridors used a 6° glide slope funnel with a 45° abort corridor and a wavefront path planner for

egress. This establishes the skid contact and airframe clearance rules used to verify physical touchdown feasibility inside HELPSs Operational Zones (Modules 4–6).

B. Monocular Deep Depth Landing Zone Classification

Rojas Perez et al. [19] developed an end to end pipeline classifying downward aerial imagery into 'Landing Zone' vs. 'No Landing Zone' without active LiDAR, using a patch based CNN generating metric depth arrays fed into an Inception derived 'LandNet' classifier trained on synthetic Gazebo depth maps and the KITTI benchmark. Depth map classification reached 82%–87% accuracy at 5 Hz, versus 67% for raw RGB serving as an onboard optical fallback if satellite elevation models are outdated or LiDAR degrades (Modules 2/5).

C. Passive Single Camera 3D Mapping and MPC Trajectory Control

Templeton et al. [20] implemented Recursive Multi Frame Planar Parallax to reconstruct 3D terrain topography from a single moving camera, storing point estimates in a multi resolution grid updated via inverse variance weighting, with 2D Integral Images enabling $O(1)$ planar fit and slope angle computation, coupled to a constrained nonlinear MPC guidance framework. This demonstrates the architectural pattern of separating $O(1)$ terrain quality evaluation from guidance dynamics that HELPSs adopts across Modules 3 and 6.

X. POLE OF INACCESSIBILITY AND GEOMETRIC CLEARANCE

A. Global Pole of Inaccessibility via Iterative Grid Subdivision

Garcia Castellanos and Lombardo [21] formalized an iterative grid subdivision algorithm to compute the Pole of Inaccessibility the point maximizing Euclidean distance to a complex polygon boundary by recursively scaling a bounding box down by $\sqrt{2}$ around the optimal candidate until millimeter level convergence, using great circle relations for spherical distances. This replaces geometric centroids, which often lie near obstacle boundaries in irregular clearings, with the PoI HELPSs uses as its Representative Point (Module 4, Step 5).

B. Largest Empty Clearance Rectangles

Chazelle et al. [22] created a divide and conquer algorithm in $O(N \log^3 N)$ time to find the largest empty rectangle among N obstacle points, combining Lower Left Voronoi diagrams, interval segment trees, and centroid decomposition to resolve boundary paths in $O(\log N)$ per query providing the mathematical foundation for extracting rectangular landing footprints for skid equipped or tandem rotor aircraft in future HELPSs versions (Modules 4/6).

C. Generalized Voronoi Diagrams and Medial Axis (VRONI)

Huber and Held [23] extended the VRONI C++ library to compute generalized Voronoi diagrams for points, segments, and circular arcs in $O(N \log N)$ expected time, using a Cone of Influence to constrain Voronoi cells and topology oriented incremental insertion. Because the PoI corresponds to a node on the polygon's medial axis, VRONI enables rapid extraction of the maximal inscribed circle candidate node i.e., the Maximum Inscribed Circle Radius HELPSs stores for every Operational Zone (Module 4, Step 6).

XI. AUTOROTATION TRAJECTORY OPTIMIZATION

Meng and Chen [24] formulated autorotation landing after engine failure as a nonlinear optimal control problem on an augmented six degree of freedom rigid body flight dynamics model, rather than the simplified two dimensional point mass models used in earlier work. Using direct transcription and a UH 60 test case, the 6 DOF solution agreed broadly with the 2 D point mass model while additionally capturing angular rates, attitudes, lateral position, and pitch control histories. Two cost formulations penalizing touchdown speed and flight time, or touchdown speed and horizontal distance produced consistent, realistic strategies even under a one second reaction delay. The safe touchdown envelope for the UH 60 case (vertical speed under 1.524 m/s, horizontal speed under 12.192 m/s) illustrates the hard dynamic constraint that any downstream site selection process must respect. HELPSs' current SDD does not yet model this envelope explicitly; it represents the clearest path to extending Module 7 (Recommendation Generation) with a dynamic feasibility filter alongside the existing geometric and slope filters.

XII. AVIATION REGULATIONS AND HELIPORT DESIGN

A. Federal Aviation Regulations (FAR) Part 91

FAR Part 91 [25] defines legal flight requirements in US airspace: minimum safe altitudes (§91.119) requiring 1,000 ft clearance over congested areas and 500 ft over non congested terrain, with explicit allowances for helicopters operating without hazard to persons or property; VFR visibility, cloud clearance, and 20 minute fuel reserve minimums (§91.155/§91.151); and

restrictions on unauthorized operation in Part 73 restricted airspace or active TFRs (§91.133), all of which HELPSs must respect when filtering candidate Operational Zones (Module 1/6).

B. ICAO Annex 14 Volume II Heliports

ICAO Annex 14 Volume II [26] establishes international heliport geometry standards: aircraft dimensioning (D value) scaling site dimensions to the critical helicopter's largest rotor turning dimension; the Final Approach and Take off Area (FATO $\geq 1.0D$ – $1.5D$) and Touchdown and Lift off Area (TLOF $\geq 0.83D$); and Obstacle Limitation Surfaces limiting FATO slope to $\leq 3\%$ for Performance Class 1 or $\leq 7\%$ ($\sim 4^\circ$) for Class 2/3 directly informing the slope thresholds used in HELPSs' Module 4 multi threshold segmentation.

C. FAA Advisory Circular AC 150/5390 2C Heliport Design

FAA AC 150/5390 2C [27] codifies an 8:1 slope VFR approach/departure surface extending 4,000 ft to a 500 ft width at 500 ft altitude, with curved approach corridors permitted when straight segment S and turn radius R satisfy $S + R \geq 1,886$ ft, and requires expanded FATO dimensions above 1,000 ft MSL to account for density altitude parameters that could refine HELPSs' Approach Corridor Score (Module 5, Step 4).

XIII. OPERABLE WEATHER AND DEGRADED VISUAL ENVIRONMENTS

A. Downwash Aerodynamics and Ground Particle Entrainment

Wadcock et al. [28] conducted full scale tuft visualization and Navier Stokes CFD modeling on UH 60L Blackhawk and EH 101 Merlin helicopters, finding peak outwash velocities and ground shear stress occur outside the rotor perimeter along stagnation lines at $y/R \approx 56\%$, with bluff fuselage geometries displacing outwash vectors relative to streamlined airframes establishing particle entrainment safety buffers for HELPSs Operational Zones with loose soil, sand, or snow (Modules 3, 5, 6).

B. Flight Path Optimization for Brownout Mitigation

Alfred [29] combined a 12 DoF aeroelastic model (HeliUM/GenHel) with a Free Vortex Method wake and Lagrangian particle tracking to optimize brownout approach trajectories, identifying two optima: a shallow approach ($\sim 5.4^\circ$) that outruns dust recirculation, and a steep approach ($\sim 9.8^\circ$) that minimizes time in ground effect dust uplift parameters HELPSs could ingest to command deceleration profiles during final descent (Modules 5/6).

C. Real Time DVE Simulation and HMD Visual Aiding

de Reus et al. [30] developed a real time pilot in the loop DVE simulation coupling FlightLab dynamics with precomputed Free Vortex Wake tables, finding that 3D scene linked conformal symbology on Helmet Mounted Displays restored spatial orientation cues during degraded visibility landings the interface architecture HELPSs could adopt to project landing trajectories onto pilot HMDs (Module 6).

XIV. RESTRICTED AIRSPACE AND COMMUNICATION CONSTRAINTS

A. Unmanned Aircraft System Traffic Management (UTM)

Kopardekar et al. [31] outlined NASA's risk based UTM Concept of Operations across Technical Capability Levels 1–4, distributing responsibility among ANSPs, USS providers, and operators via real time SWIM API feeds for airspace restrictions, geofences, and NOTAMs connecting HELPSs to live geofence/TFR queries prior to site selection (Modules 1/6).

B. Communication Aware Trajectory Planning in Urban Environments

Oh et al. [32] developed a 3D communication aware trajectory planner for urban UAVs, constructing a Communication Feasible Area from 2D radial visibility cross sections, a Gaussian Process Regression model for loitering orbit selection, and Decentralized NMPC guaranteeing HELPSs' chosen landing points maintain unbroken RF line of sight with ground support in urban obstacle fields (Modules 2, 5, 6).

C. Federal Special Use Airspace (SUA) Regulation

FAA Order JO 7400.10F [33] establishes statutory boundaries for Restricted Areas (§73.11–§73.72), Prohibited Areas (§73.81–§73.94), and non regulatory MOAs, Alert Areas, and Warning Areas ingested by HELPSs to filter prohibited flight zones during real time site selection (Modules 1/6).

XV. SYNTHESIS AGAINST THE HELPSS SYSTEM DESIGN

Read together, the thirty three papers reviewed decompose cleanly onto the eight modules of the HELPSSs Terrain Knowledge Generation Pipeline and the seven modules of the Helicopter Landing Recommendation Pipeline described in the SDD. Table I maps each literature domain to its supporting module and reproduces the must read / important priority flags used during the HELPSSs literature triage.

TABLE I
LITERATURE TO MODULE MAPPING

Literature Domain	Papers	Primary HELPSSs Module / Phase	Priority
Object Detection	[1]–[3]	Module 2: Candidate Terrain Identification	Must read [3]; support [2]
Land Cover Classification	[4]–[6]	Module 2/3: Terrain Classification & Characterization	Important
Connected Component Analysis	[7]–[9]	Module 2 & 4: Terrain/Zone Formation; Phase 8 Optimization & Scalability	Must read [7]; important [8]; roadmap [9]
Macro Landform Classification	[10]–[11]	Module 3: Terrain Characterization	Important
Terrain Representation	[12]–[13]	Module 4/5 & Terrain Knowledge Database (Ch. 5)	Important
DEM Construction & Error	[14]–[15]	Module 1: Data Acquisition; Module 4: Slope Computation	Important
Slope & Aspect Derivation	[16]–[17]	Module 4: Slope Analysis & Operational Zone Generation	Important
Landing Zone Detection & Validation	[18]–[20]	Module 4 & 7: Slope/Footprint Validation & Recommendation	Must read [18]; roadmap [19]–[20]
Pole of Inaccessibility / Geometry	[21]–[23]	Module 4 & 7: Operational Zone Generation; Landing Footprint Validation	Must read [21], [23]; roadmap [22]
Autorotation Trajectory Optimization	[24]	Module 7: Recommendation Generation (feasibility filter gap)	Not yet integrated
Aviation Regulations & Heliport Design	[25]–[27]	Module 1/6: Restricted Area Exclusion; Slope thresholds	Foundational / regulatory
Operable Weather / DVE	[28]–[30]	Module 3, 5, 6: Environmental Context; Weather & Wind Analysis	Important
Restricted Airspace & Communications	[31]–[33]	Module 1/6: Restricted Area Exclusion; Nearby Terrain Retrieval	Foundational / regulatory

Two research gaps stand out once the combined literature is checked against the SDD. First, the Pole of Inaccessibility and rectangle fitting algorithms reviewed in Section X are restricted to axis aligned or 2D unweighted domains, whereas real Operational Zones are irregular polygons on sloped, unevenly weighted terrain – consistent with the gap already flagged in the original HELPSSs review. Second, and newly surfaced by merging in the autorotation literature, none of the thirty three papers couples terrain and land cover perception output to an autorotation dynamics optimal controller: the optimal control work in Section XI assumes the landing point and local terrain are already known, while the terrain perception work in Sections II–X does not consider aircraft dynamics or reaction time constraints. HELPSSs Version 1's Recommendation Generation module (Module 7) filters on slope, footprint, and mission context, but does not yet verify that the candidate Operational Zone lies within the helicopter's dynamically reachable autorotation envelope a natural Phase 8+ extension.

XVI. CONCLUSION

This combined review demonstrates that reliable autonomous emergency landing for rotorcraft requires integrating several distinct computational domains that were previously documented in two separate literatures. Macro scale satellite and land cover processing [2]–[6] provides candidate operational areas that feed deterministic or parallelized connected component analysis [7]–[9] to isolate contiguous clear terrain, contextualized against macro landform classification [10]–[11] and represented in morphology preserving structures [12]–[13] built atop DEMs whose error characteristics are well understood [14]–[15] and whose slope derivatives carry known, resolution dependent biases [16]–[17]. These regions are processed using spatial clearance models [21]–[23] and verified against 3D physical skid fitting algorithms [18]–[20], while candidate touchdown coordinates are evaluated against regulatory airspace boundaries [25]–[27], [33], communication line of sight constraints [32], and aerodynamic downwash hazards [28]–[30]. The autorotation dynamics literature [24] is mature in isolation but remains unconnected to any of the terrain perception literature reviewed here, defining the primary open direction for HELPSs beyond Version 1.

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